

CO5 ASSESSMENT TOOL-1

CONSTRAINT -BASED PROBLEM SOLVING

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1.Real-Time Face Recognition for Campus Security

Introduction

A real-time face recognition system uses a CCTV camera to identify registered students and staff at the university entrance. The system processes live video and compares detected faces with a registered face database.

Pipeline:

Image Acquisition → Preprocessing → Face Detection → Feature Extraction → Recognition
→ Verification → Identification

Image Acquisition

- Capture live video using a CCTV camera or webcam.
- OpenCV VideoCapture() is used to read video frames. A resolution such as 640×480 can be used for faster processing.

Preprocessing

Preprocessing improves image quality and prepares the face for recognition.

- Resize the image to reduce processing time.
- Convert the image to grayscale. Remove noise using Gaussian filtering.
- Use CLAHE or histogram equalization to improve faces under different lighting conditions.

Face Detection

The system detects faces in every video frame.

For a low-cost computer, Haar Cascade Classifier can be used with OpenCV.

```
faces = face_cascade.detectMultiScale(  
    gray, scaleFactor=1.1, minNeighbors=5  
    )
```

The detected face is then cropped from the image.

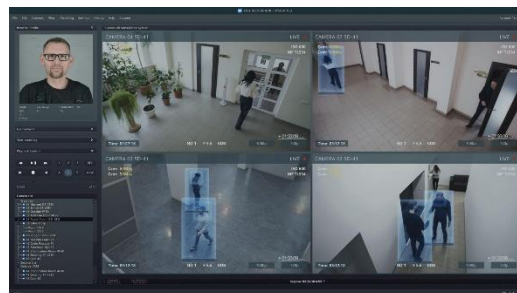
Face Alignment

After detecting the face, the system performs face alignment to obtain a properly positioned facial image. Alignment adjusts the face according to important facial landmarks such as the eyes, nose, and mouth.

The aligned face is then cropped and resized to a fixed dimension before feature extraction. This reduces the effect of different head positions, face angles, camera distances, and slight rotations.

Benefits of face alignment:

- Improves consistency between face images. Reduces the effect of head rotation.
- Makes feature extraction more accurate. Improves the reliability of face recognition.



Feature Extraction

Feature extraction identifies the important characteristics of the detected face and converts them into a form that can be compared with registered faces.

For a lightweight OpenCV system, **LBPH (Local Binary Patterns Histogram)** can be used. LBPH analyzes the local texture patterns of different regions of the face and creates a numerical representation.

Important facial regions include:

- Eyes ,Nose ,Mouth ,Cheeks,Overall facial texture

The extracted features from the live camera image are compared with the features stored in the face database.

For higher accuracy, deep face embeddings can also be used. These represent each face as a numerical vector and provide better performance for changes in pose and lighting.

Face Database

A face database is maintained to store the facial information of registered students and staff members. During the registration process, several face images are collected for each individual.

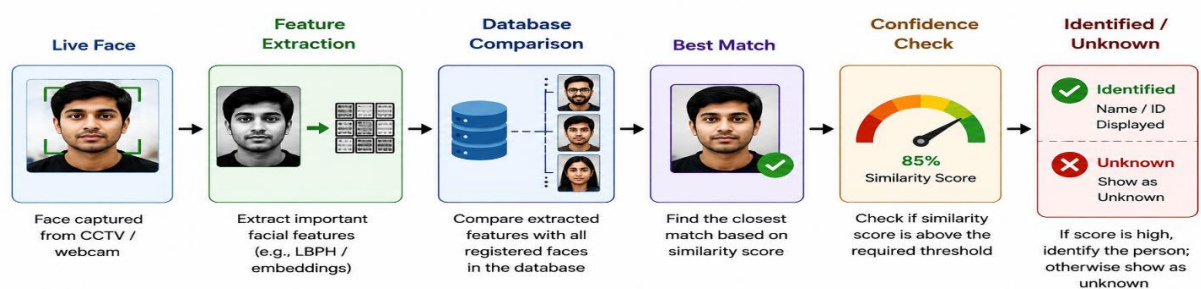
The database can contain information such as:

ID	Name	Face Features
101	Student A	Registered features
102	Student B	Registered features
103	Staff A	Registered features

Multiple samples should be collected under different:

- Lighting conditions ,Facial expressions ,Head positions,Camera distances

Face Recognition:



Reducing False Recognition

False recognition can be minimized by:

- Using multiple training images. Applying proper preprocessing.
- Setting an appropriate confidence threshold. Using face alignment.

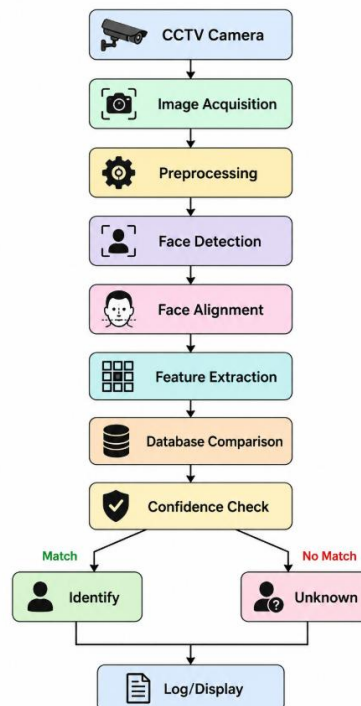
- Confirming the same identity across multiple frames. Rejecting low-confidence matches.

Real-Time Optimization

To run efficiently on a low-cost computer:

- Use 640×480 resolution.
- Process every 2nd or 3rd frame instead of every frame.
- Process only detected face regions.
- Use lightweight algorithms such as Haar Cascade and LBPH.
- Restrict detection to the entrance region when possible.

Complete Pipeline



Result: The proposed OpenCV-based system provides a practical solution for campus security. Preprocessing and alignment improve accuracy, while confidence checking reduces false recognition. Frame skipping, reduced resolution, and lightweight algorithms allow the system to maintain real-time performance on a low-cost computer.

2.Intelligent Traffic Violation Detection System

Introduction

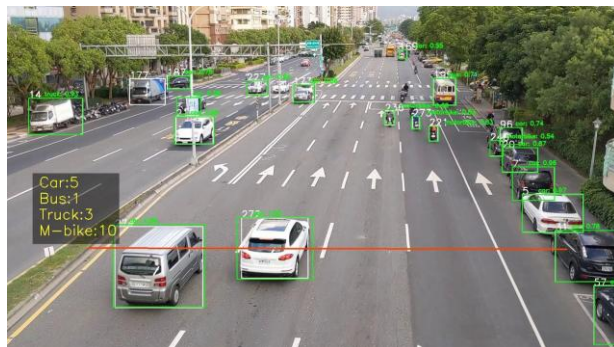
An Intelligent Traffic Violation Detection System uses roadside cameras to monitor vehicles automatically. The system captures live traffic video, detects vehicles, tracks their movement, and counts vehicles when they cross a predefined line.

Pipeline:

Video Acquisition → Preprocessing → Vehicle Detection → Motion Analysis → Vehicle Tracking → Line Crossing → Violation Detection → Counting → Visualization

Video Acquisition

- Capture live traffic video using a roadside CCTV camera. OpenCV VideoCapture() is used to read video frames.
- A suitable resolution such as 640×480 can be used for real-time processing.
- The camera should be positioned to clearly view the road and counting line.



Preprocessing

Preprocessing improves the quality of traffic video before vehicle detection.

- Resize frames to reduce processing time.
- Convert frames to grayscale when required. Apply Gaussian or median filtering to reduce camera noise.
- Use contrast enhancement to improve visibility. Apply background subtraction for detecting moving vehicles.

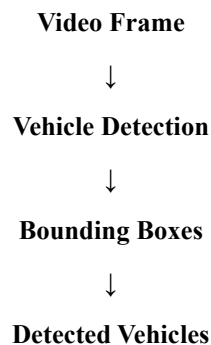
These steps help the system work better under different environmental conditions.

Vehicle Detection

The system identifies vehicles such as: Cars ,Buses ,Trucks .Motorcycles

OpenCV can use background subtraction methods such as MOG2 or KNN for basic traffic monitoring.

For improved accuracy, an object detection model such as YOLO can be integrated with OpenCV.



Motion Analysis

Motion analysis determines whether detected objects are actually moving vehicles.

Background subtraction separates moving objects from the static road background.

The system can calculate the movement of each detected vehicle between consecutive frames.

For example:

Frame 1 → Vehicle at Position A

Frame 2 → Vehicle at Position B

The change in position indicates the direction and movement of the vehicle.

Vehicle Tracking

After detection, each vehicle is assigned a unique tracking ID.

Example:

Vehicle 1 → ID 01

Vehicle 2 → ID 02

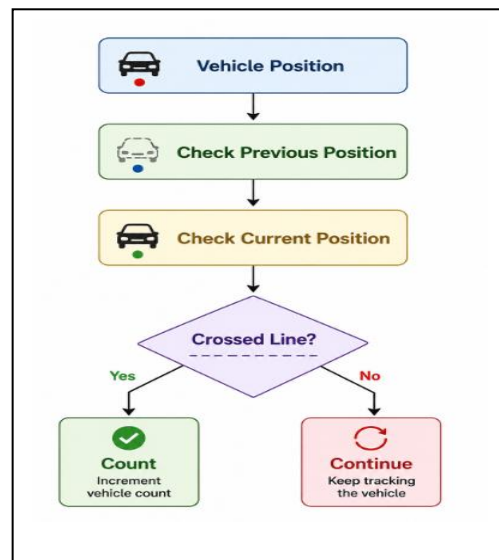
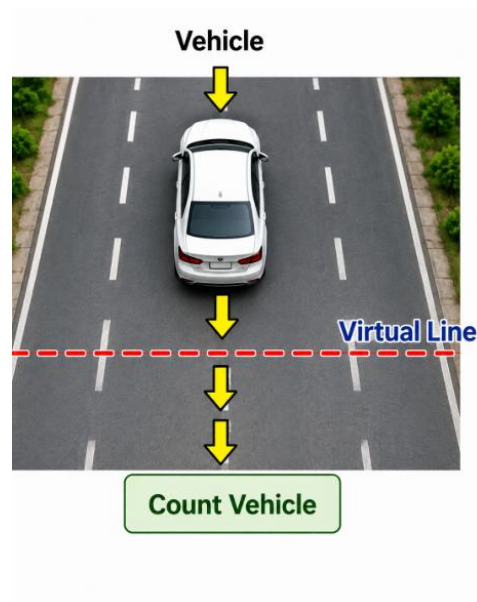
Vehicle 3 → ID 03

Tracking algorithms such as Centroid Tracking, CSRT, or SORT can be used.

Tracking prevents the same vehicle from being counted repeatedly in different frames.

Line-Crossing Detection

A virtual line is drawn across the road.



Vehicle Counting

Each tracked vehicle is counted only once using its unique tracking ID.

Vehicle ID	Direction	Status
01	Forward	Counted
02	Forward	Counted
03	Reverse	Alert

Example: Cars = 25, Buses = 5, Trucks = 8, Total = 38

Abnormal Movement Detection

The system analyzes vehicle movement to identify:

- Wrong-direction movement.Sudden stopping ,U-turns
- Unusual lane changes ,Restricted-area entry

Detected violations generate real-time alerts.

Handling Shadows and Camera Noise

Shadows

- Use shadow removal in background subtraction.
- Apply morphological operations. Analyze object size and shape.

Camera Noise

- Use Gaussian or median filtering. Apply image normalization.

These techniques reduce false detections.

Handling Dynamic Backgrounds

Moving trees, rain, and changing sunlight can cause false detections.

- Adaptive background subtraction ,Morphological filtering
- Region of Interest (ROI) ,Object detection when necessary

ROI ensures that only the road area is analyzed.

Handling Multiple Vehicles

Each vehicle receives a unique tracking ID.

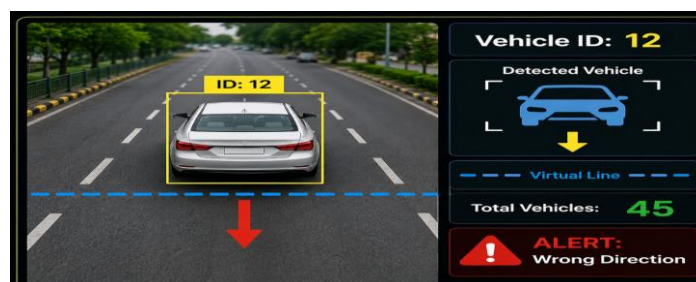
Vehicle 01 → ID 01 Vehicle 02 → ID 02 Vehicle 03 → ID 03 Vehicle 04 → ID 04

Tracking maintains vehicle identity across frames and prevents duplicate counting.

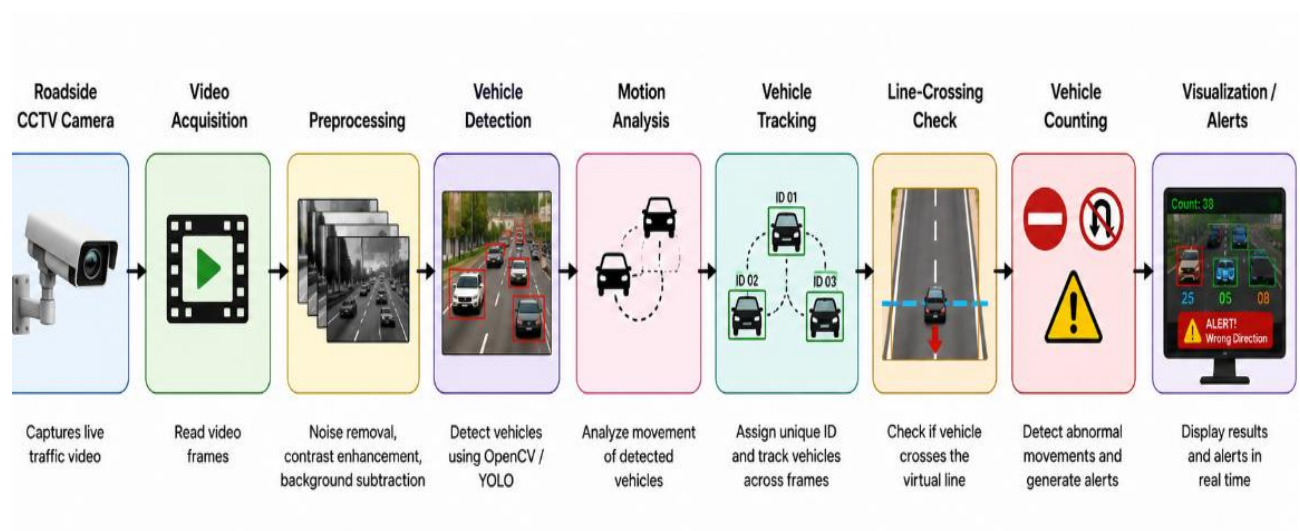
Real-Time Visualization

The output screen displays:

- Vehicle bounding boxes ,Tracking IDs ,Virtual counting line ,Vehicle count ,Movement direction ,Violation alerts



Complete System Pipeline



Real-Time Optimization

To improve performance:

- Use 640×480 resolution.
- Process selected frames. Restrict detection to the road ROI.
- Use lightweight detection and tracking. Avoid repeated detection of already tracked vehicles.

Result

The OpenCV-based system can detect, track, count, and monitor vehicles in real time. Line-crossing detection provides accurate counting, while motion analysis identifies traffic violations. Filtering, ROI selection, and tracking help reduce false detections and duplicate counting.

3. Automated Document Processing System

Introduction

An Automated Document Processing System uses OpenCV to improve scanned documents and prepare them for OCR (Optical Character Recognition). It removes noise, corrects document geometry, segments text, and produces clear text efficiently.

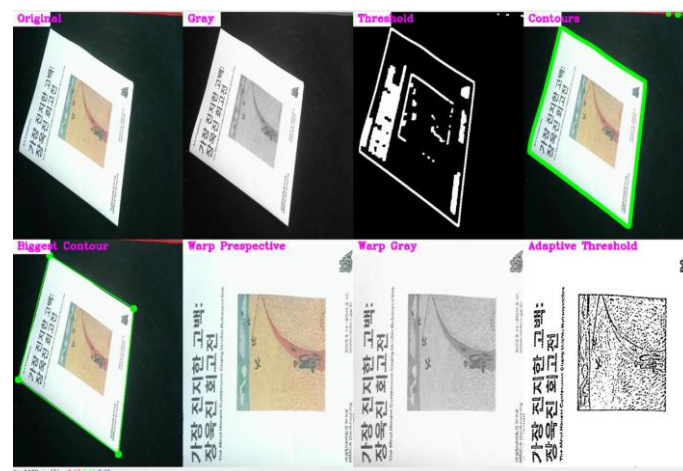
Pipeline:

Image Acquisition → Grayscale → Filtering → Thresholding → Morphological Processing → Segmentation → Geometric Correction → OCR

Grayscale Conversion

The scanned image is converted from color to grayscale.

- Reduces image complexity.
- Removes unnecessary color information.
- Makes text and background easier to distinguish.
- Improves the efficiency of further processing.
- `gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)`



Filtering

Filtering removes unwanted noise such as dots, scratches, and scanning artifacts.

Common methods include:

- Gaussian filter – reduces general noise.
- Median filter – removes salt-and-pepper noise.

Cleaner images help OCR recognize characters more accurately.

Thresholding

Thresholding separates text from the background.

Methods:

- Global thresholding
- Adaptive thresholding

- Otsu thresholding

Adaptive thresholding is useful for documents with uneven illumination because it calculates the threshold locally.

Morphological Operations

Morphological operations improve the structure of text.

- Opening: removes small unwanted objects and noise.
- Closing: fills small gaps in characters.
- Dilation: strengthens thin text.
- Erosion: removes excess pixels.

These operations produce clearer characters for OCR.

Text Segmentation

The processed document is divided into meaningful regions such as:

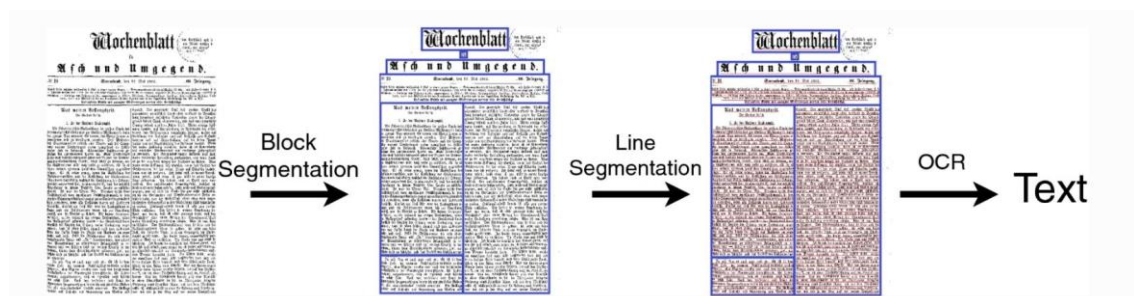
- Paragraphs, Lines, Words, Tables

Contours and connected components can be used to identify text regions. Segmentation prevents nearby text from interfering with recognition.

Geometric Correction

Scanned documents may be tilted or distorted. Deskewing and perspective correction are applied to straighten the document.

This ensures that text lines are properly aligned and improves OCR accuracy.



OCR and Batch Processing

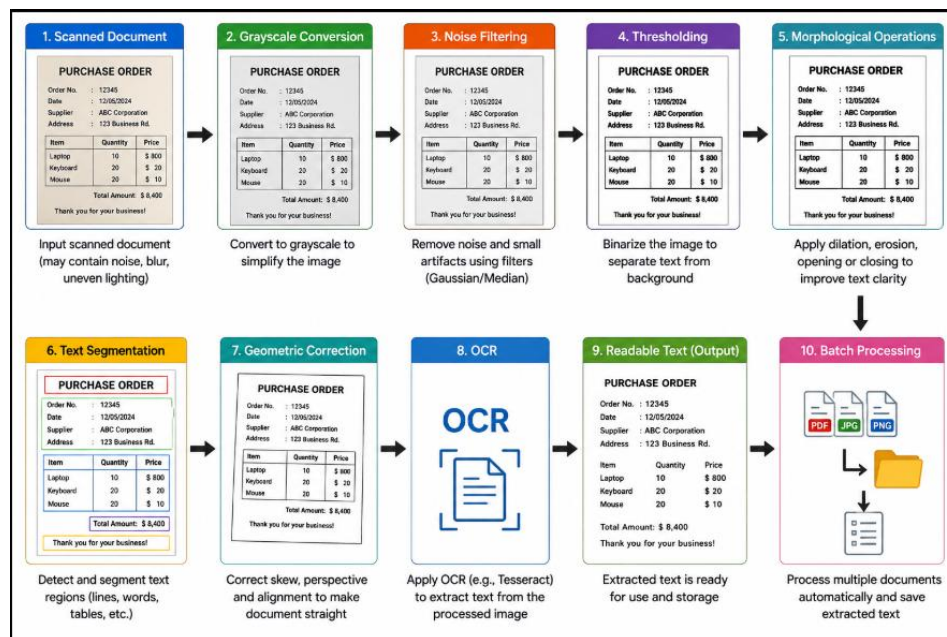
After preprocessing, the clean image is passed to an OCR engine such as Tesseract.

For large document collections, batch processing can automatically process files one by one:

Input Folder → Preprocessing → OCR → Text Output

This reduces manual effort and allows thousands of documents to be processed efficiently.

Complete Pipeline:



Result:

The OpenCV-based pipeline improves OCR accuracy by removing noise, separating text from the background, repairing characters, segmenting text regions, and correcting document alignment. Batch processing allows large numbers of scanned documents to be processed quickly and consistently.

4.Real-Time Multi-Person Tracking and Attendance System

Introduction

A Real-Time Multi-Person Tracking and Attendance System uses classroom CCTV cameras to automatically identify students and record attendance. The system detects multiple faces, recognizes registered students, tracks them across frames, and marks attendance only once.

Pipeline:

CCTV Camera → Preprocessing → Face Detection → Feature Extraction → Face Recognition → Tracking → Attendance Verification → Attendance Record

Video Acquisition

- Capture live classroom video using CCTV cameras.
- Use OpenCV VideoCapture() to read frames.
- A resolution such as 640×480 can be used for faster processing.
- The camera should cover the entire classroom.

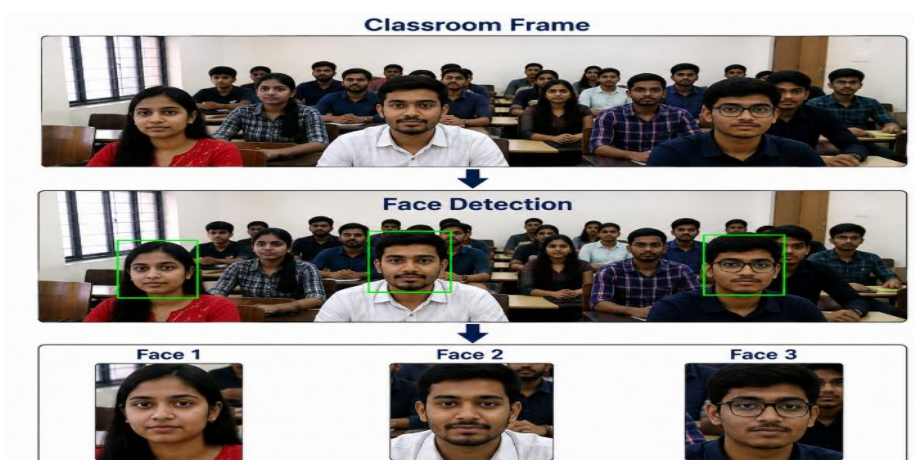
Preprocessing

Preprocessing improves the quality of the captured faces.

- Convert frames to grayscale. Resize frames to reduce processing time.
- Apply Gaussian or median filtering to remove noise.
- Use histogram equalization or CLAHE for uneven lighting. .

Face Detection

OpenCV methods such as Haar Cascade or DNN-based face detection can be used.



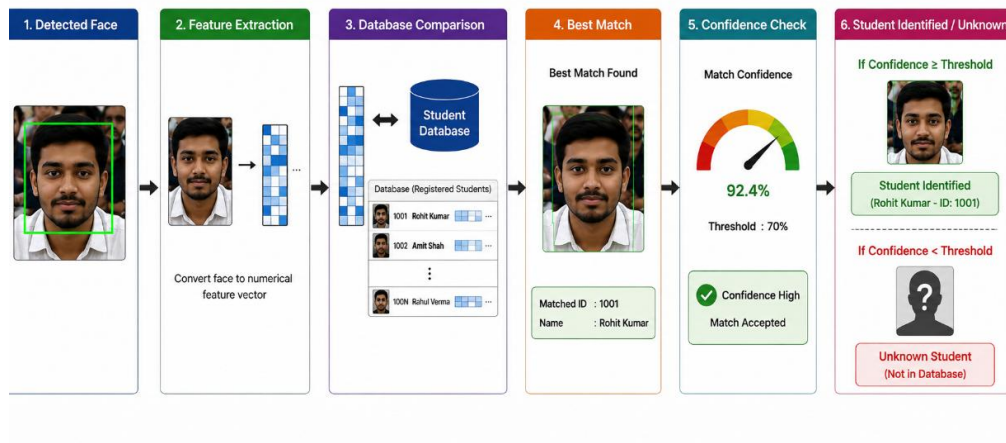
Feature Extraction

Important facial features are extracted from each detected face.

For a lightweight system, LBPH (Local Binary Patterns Histogram) can be used. For better accuracy, deep face embeddings can be used.

The extracted features are compared with the registered student database.

Face Recognition



Multi-Person Tracking

After recognition, each person is given a unique tracking ID.

Student A → Track ID 01

Student B → Track ID 02

Student C → Track ID 03

Tracking algorithms such as CSRT, SORT, or centroid tracking can maintain the identity of students across consecutive frames.

This prevents the system from treating the same student as a new person in every frame.

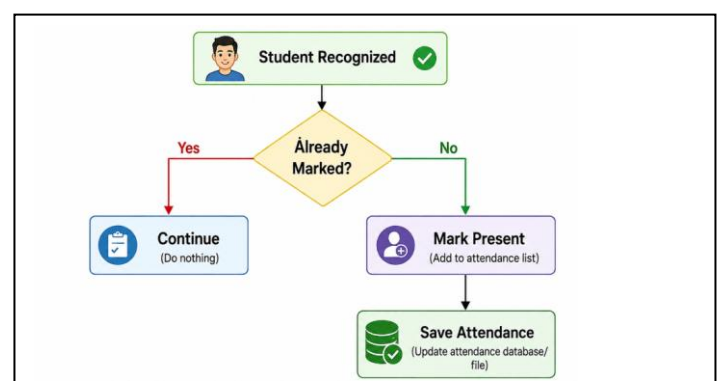
Attendance Management

When a registered student is recognized, the system checks the attendance list.

Once attendance is marked, the student's ID is stored in a set or database. The same student is not marked again.

Example:

Student ID	Name	Attendance
101	Student A	Present
102	Student B	Present
103	Student C	Absent



Handling Temporary Face Disappearance

A student may temporarily turn away, move their head, or become blocked by another student.

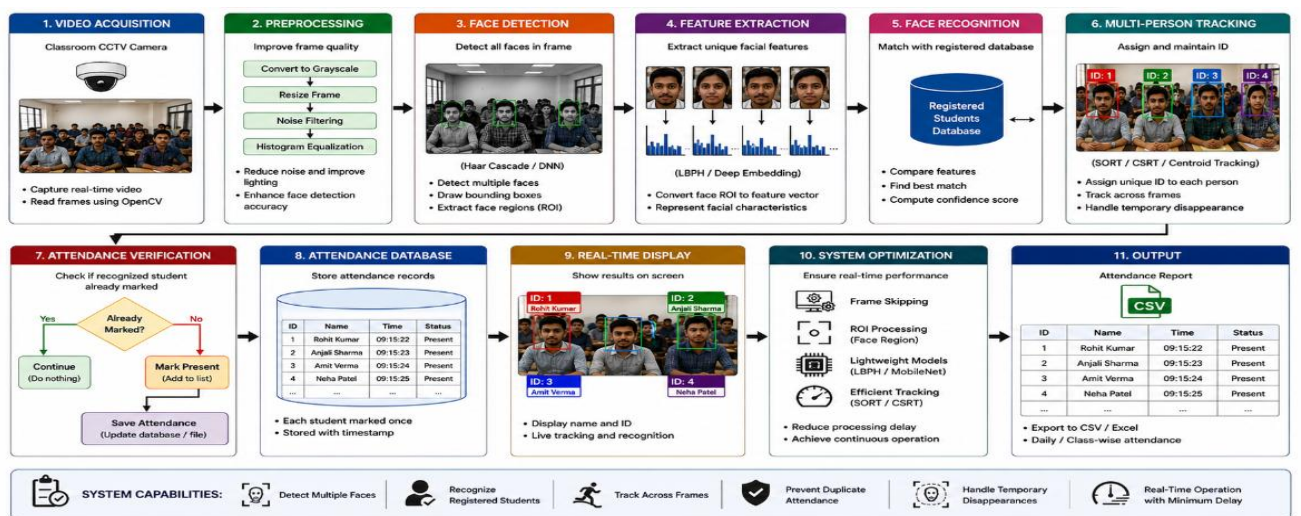
The tracking system can maintain the student's tracking ID for a short period even when the

Real-Time Optimization

To reduce processing delay:

- Use 640×480 resolution. Process selected frames instead of every frame.
- Use lightweight face detection and tracking algorithms. Track detected students between detection frames.
- Process only face regions instead of the complete image. Use ROI to focus on the classroom area.

Complete Pipeline:



Result

The proposed system can automatically detect, recognize, track, and record multiple students from classroom CCTV video. Unique tracking IDs prevent repeated identification, while the attendance database ensures that each student is marked only once. Temporary face disappearance can be handled through tracking, and frame skipping and lightweight algorithms help maintain continuous real-time operation with minimum processing delay.

5. Smart Home Surveillance and Intrusion Detection

Introduction

A Smart Home Surveillance System uses a fixed camera to continuously monitor an entrance and detect suspicious movement. OpenCV processes the video, identifies moving objects, tracks them, and generates an alert when an unauthorized person enters a restricted region.

Pipeline:

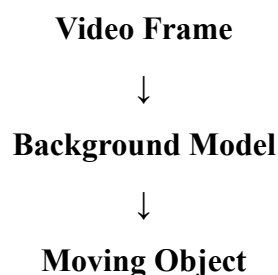
Camera → Preprocessing → Background Subtraction → Filtering → Morphological Operations → Contour Detection → Motion Analysis → Tracking → Intrusion Detection → Alert

Video Acquisition

- Capture live video using a fixed CCTV/webcam.
- Use OpenCV VideoCapture() to read frames. Use a resolution such as 640×480 for real-time performance.

Background Subtraction

Background subtraction separates moving objects from the stationary background. Methods such as MOG2 or KNN can be used.



This helps detect people entering the monitored area.

Filtering and Morphological Operations

Noise and small unwanted movements are removed using:

- Gaussian/Median filtering – reduces camera noise.
- Opening – removes small objects.
- Closing – fills gaps in detected objects.

- Dilation – strengthens object regions.

These operations produce cleaner motion masks and reduce false detections.

Contour and Motion Detection

Contours are used to identify the boundaries of moving objects.

Small contours caused by leaves, insects, or camera noise can be ignored using an area threshold.

Motion is analyzed by comparing object positions between consecutive frames.

Object Tracking

Each detected moving object is assigned a unique tracking ID.

Person → ID 01

Person → ID 02

Simple centroid tracking or lightweight trackers such as CSRT can be used to follow objects across frames.

Intrusion Detection

A Region of Interest (ROI) is defined around the entrance.



Reducing False Alarms

Shadows

- Use shadow removal with MOG2.
- Apply morphological operations.
- Check object shape and area.

Illumination Changes

- Use adaptive background models. Apply image normalization.

Trees and Background Movement

- Ignore very small contours. Define an entrance ROI.
- Use object size and motion direction.

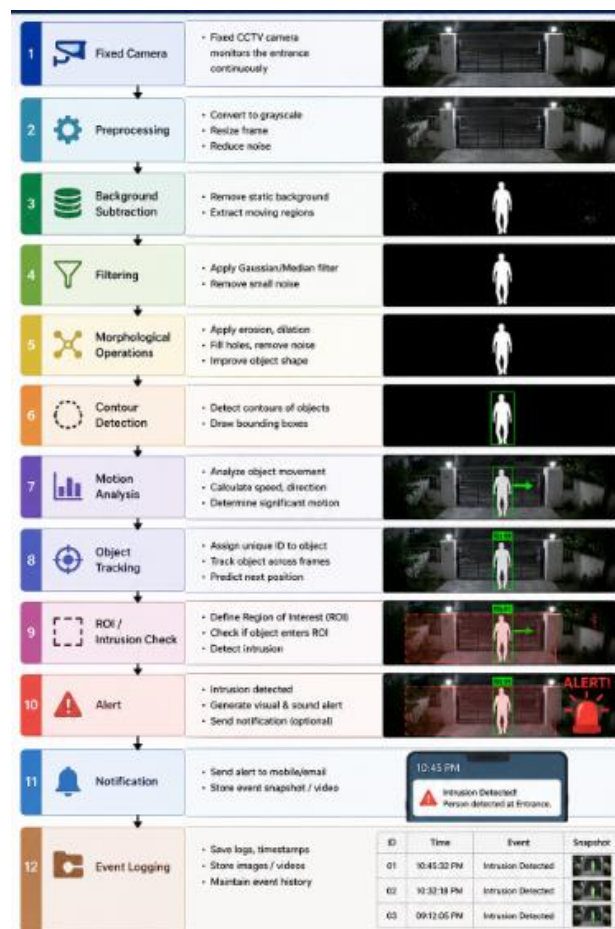
These techniques help distinguish real human movement from unwanted background changes.

Real-Time Optimization

For limited hardware:

- Use 640×480 resolution. Process selected frames. Restrict processing to the ROI.
- Use lightweight MOG2/KNN and contour detection. Track objects instead of detecting them repeatedly.

Complete Pipeline



Result:The proposed OpenCV system can detect and track significant movement while reducing false alarms caused by noise, shadows, lighting changes, trees, and small background movements. ROI-based intrusion detection and lightweight processing allow the system to provide real-time alerts on low-cost hardware.